

Original papers

On field recognition of pig cough based on multimodal audio representation and fusion with contrastive learning

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ARTICLE INFO

Keywords:

Multimodal pig audio
Cough recognition
Contrastive learning
Self-supervised learning
Pig respiratory disease

ABSTRACT

Cough is an early clinical sign of pig respiratory diseases, of which accurate and timely recognition is critical to early warning of pig respiratory diseases. Recent pig cough recognition methods suffered from limited label training samples and failed to obtain powerful audio representations. In this study, a novel multimodal pig audio representation and fusion (MSS) framework was proposed to learn meaningful feature representations. Based on the MSS framework, this study proposed a method for recognizing pig cough in practical production conditions. The results showed that the MSS framework demonstrated excellent capability of learning pig audio representations, based on which the pig cough recognition method achieved accurate results by fusing the mel-frequency cepstral coefficient (MFCC) and log-mel spectrogram, with an accuracy of 91.37 % using a simple linear classifier. After fine-tuning the supervision dataset, the performance of the pig cough recognition method was further improved to 94.32 % and outperformed the state-of-the-art supervised pig cough recognition methods. The proposed MSS framework effectively addressed the issue that the labeled training samples were limited in practical production conditions, enabling the proposed pig cough recognition method to maintain accuracy even with a 50 % reduction in training samples and to yield decent recognition results with approximately 3 % of the total cough samples. This study indicates that the proposed pig cough recognition method based on the MSS framework can effectively complement the supervised methods and make a promising tool for early warning of pig respiratory diseases.

1. Introduction

Respiratory disease is a global threat to contemporary pig production (Boeters et al., 2023; Holtkamp et al., 2013; Pessoa et al., 2021; Zhang et al., 2022), causing extensive economic losses by increasing mortality, reducing the growth rate, and lowering the feed conversion efficiency (Boeters et al., 2023; Ferrari et al., 2008; Silva et al., 2008; Upadhyaya et al., 2023b). It is estimated that pig respiratory disease costs \$664 million annually in the United States (Holtkamp et al., 2013) and about \$200 per sow in China (Zhang et al., 2022). Therefore, initiating early treatment of pig respiratory disease is of great significance to pig performance and welfare (Pessoa et al., 2021). As an early clinical sign of pig respiratory diseases (Guarino et al., 2008; Racewicz et al., 2021; Wang, 2024; Xie et al., 2024; Yin et al., 2021), number of coughs can objectively indicate the health of pig respiratory (Berckmans et al., 2015; Guarino et al., 2008; Olczak et al., 2023). However, manual

measurement of pig cough is time-consuming and can be biased (Pessoa et al., 2021; Shen et al., 2022), especially on large pig farms, delaying intervention strategies to mitigate productivity losses. Therefore, there is a growing demand for the automatic and reliable measurement of pig coughs.

Recently, cutting-edge deep learning (DL) technology has been extensively adopted for analyzing pig audio signals, significantly advancing the automatic recognition of pig coughs. Accordingly, many pig cough recognition methods have been proposed based on DL models, such as convolution neural networks (CNNs) (Liao et al., 2022; Pann et al., 2024; Shen et al., 2021; Song et al., 2022; Upadhyaya et al., 2023a); Yin et al., 2021) and the Audio Spectrogram Transformer (AST) (Xie et al., 2024), and achieved remarkable progress (Ji et al., 2022; Shen et al., 2022; Wang, 2024; Xie et al., 2024; Yin et al., 2023, 2021). Wherein, one of the most commonly used procedures is to convert audio waveforms to spectral representations as the model input, such as the

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<https://doi.org/10.1016/j.compag.2025.110842>

Received 17 March 2025; Received in revised form 31 July 2025; Accepted 4 August 2025

Available online 9 August 2025

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mel-frequency cepstral coefficient (MFCC) (Ji et al., 2022; Liao et al., 2022; Pan et al., 2024; Shen et al., 2022, 2021; Song et al., 2022; Wang et al., 2019; Yin et al., 2023; Zhang et al., 2019) and log-mel spectrograms (Upadhyaya et al., 2023a; Xie et al., 2024). Alternatively, features extracted from audio waveforms were directly used to recognize the pig coughs (Hou et al., 2024; Ji et al., 2022; Pan et al., 2024; Wang et al., 2019). In field applications, the acoustic environment in a pig barn is very complicated (Shen et al., 2022; Wang, 2024; Yin et al., 2023), which is full of noises, similar and overlapping pig vocalizations, posing significant challenges for pig cough recognition. Therefore, single audio modality may not be sufficient in practical production conditions, and fusing multimodal audio features for pig cough recognition has been proposed. Ji et al. (2022) proposed to combine acoustic features extracted from the audio waveforms and visual features extracted from the corresponding spectrograms for pig cough recognition. The authors demonstrated that the concatenation of acoustic and visual features improved the model performance compared to each single feature modality. Similar results were also reported in Shen et al. (2022) and Yin et al. (2023), in which the visual features were extracted by a pre-trained CNN instead of by manual extraction, resulting in further improved accuracy of pig cough recognition. Although these studies achieved promising results, they are still subjected to following issues. (1) the accurate recognition of pig coughs highly depends on the features extracted from the pig audio (Ji et al., 2022; Kim and Choi, 2023; Laguna et al., 2023; Shen et al., 2022). However, previous studies (Kim and Choi, 2023; Shen et al., 2022, 2021; Yin et al., 2023, 2021) partially failed to obtain powerful audio representations due to the manual feature extraction or limited labeled training samples. Besides, state-of-the-art DL models relied heavily on a substantial number of labeled samples, which was challenging to collect in practical production conditions (Ma et al., 2018; Upadhyaya et al., 2023b), (2) these methods simply concatenated the multimodal audio features, ignoring the fact that the multimodal audio features characterized an audio sample from varying perspectives (Ji et al., 2022), which should not be indiscriminately stacked in different applications, such as different breeds, ages and barns (Ma et al., 2023; Shen et al., 2022; Upadhyaya et al., 2023a; Upadhyaya et al., 2023b), (3) few studies focused on learning feature representations directly from the raw waveforms. Besides, the fusion of the high-level semantic features derived from the spectral representations and raw audio waveforms to enhance the representation ability and adapt to variations was yet to be explored. Therefore, more studies are still needed to facilitate multimodal audio representation and fusion for pig cough recognition in practical production conditions.

Self-supervised learning (SSL) has emerged as a dominating approach to leverage large amounts of unlabeled data to learn powerful feature representations (Kim and Choi, 2023; Niizumi et al., 2023; Saeed et al., 2021; Wang et al., 2021; Wang and van den Oord 2021). Among the self-supervised learning methods, contrastive learning is particularly effective and computationally efficient at learning meaningful representations by discriminating positive pairs from negative pairs (Li et al., 2024; Saeed et al., 2021). Recently, contrastive learning has been successfully transferred from the computer vision domain to the audio domain. For example, the bootstrap your own latent (BYOL) framework (Grill et al., 2020), which attracts great interest since it does not require negative pairs during the training process and achieves good accuracy in downstream tasks, has been generalized to the SSL for audio representations, such as BYOL-A (Niizumi et al., 2023) and WaveBYOL (Kim and Choi, 2023). However, contrastive learning has been primarily applied to general-purpose audio representations, and its performance on pig audios collected in practical conditions remains unclear. In addition, the widely used contrastive learning methods generally involve siamese networks in favor of comparison, which cannot be directly applied to multimodal audio representation and fusion, such as the raw waveform and spectrogram modalities.

In order to tackle the above challenges, this study proposed a novel method to accurately and reliably recognize pig coughs in practical

production conditions. Specifically, a multimodal pig audio representation and fusion framework based on contrastive learning (MSS) was proposed, which learned feature representations from multimodal pig audio using two different networks. Consequently, each network could build a representation for its respective input modality via the supervision provided by the other one. The learned multimodal feature representations were used for the follow-up pig cough recognition. The objectives of this study were (1) to investigate the potential of the proposed multimodal pig audio representation and fusion framework with contrastive learning, (2) to evaluate the performance of the proposed pig cough recognition method in practical production conditions, (3) to verify if the proposed multimodal pig audio representation and fusion framework was able to address the issue of limited training samples.

2. Materials and methods

2.1. Experimental setup and dataset construction

The pig vocalization data was collected in June 2023 at barn No.3 in the Wutao experimental farm of Chongqing Academy of Animal Sciences, Rongchang, China. The barn had a length of approximately 60 m and a width of 9 m, consisting of 48 pens (Fig. 1). Pen No. 10 was adopted as the experimental pen since manual inspections found that coughing had been occurring for several days. The experimental pen (Fig. 1, marked in red) housed six growing pigs (around 30 kg) and had an area of 10.8 m² with a half-slatted concrete floor. The pig audio recording device used in this study was a Zoom H2n stereo recorder with five built-in microphones (Zoom corporation, Tokyo, Japan), which were mounted above the experiment pen at a distance of 1.5 m to the ground. The collected pig audio was 16-bit with a sampling rate of 44.1 k Hz and was stored in WAV format.

Since the pig audio data were collected in practical production conditions, the background noise mainly included fans for ventilation and metal collisions. In order to enhance the model ability to adapt to the complicated acoustic environment in a pig barn, as well as to improve the applicability in the practical production conditions, this study did not adopt quality control or data denoising procedures. After data collection, the audio recordings were down sampled to 16 k Hz, aiming to reduce the computing costs, as well as to speed up the training processing. In the next, this study built two datasets, including the pre-training and the supervision datasets with each containing half of the collected pig audio data. Notably, the raw audio data was randomly assigned into the pre-training and supervision datasets, and there was no data overlap between the two datasets. The pre-training dataset was used for the construction of the proposed MSS, enabling the transfer of learned knowledge to the downstream task of pig cough recognition. In order to build the pre-training dataset, this study employed a sliding window to extract segments from the pig sound recordings as model input since the original pig sound recordings (about three hours per recording) could not be directly fed to the networks. Specifically, the sliding window had a length of 1.5 s and a stride of 1.5 s, thus extracting 40,548 fixed-length segments without overlap between two adjacent segments. Subsequently, the pre-training dataset was divided into two subsets, i.e., the training and the validation datasets containing 32,433 and 8115 segments, respectively.

Regarding the supervision dataset, this study used Adobe Audition 2023 to label the pig vocalizations. With the assistance of an experienced veterinarian, four pig vocalization categories, including cough, scream, sneezing, and other pig sounds (Fig. 2), were labeled for dataset construction. Notably, vocalization samples that were heavily obscured by noise would be excluded during the data labeling. The number of samples for the four vocalization categories was 3090, 3930, 1110, and 1370, respectively. Considering the different durations of sound events, the vocalization samples with duration less than 1s were filled with zeros to 1s, while those with durations more than 1s were randomly cropped to 1s. In the following step, the supervision dataset was further

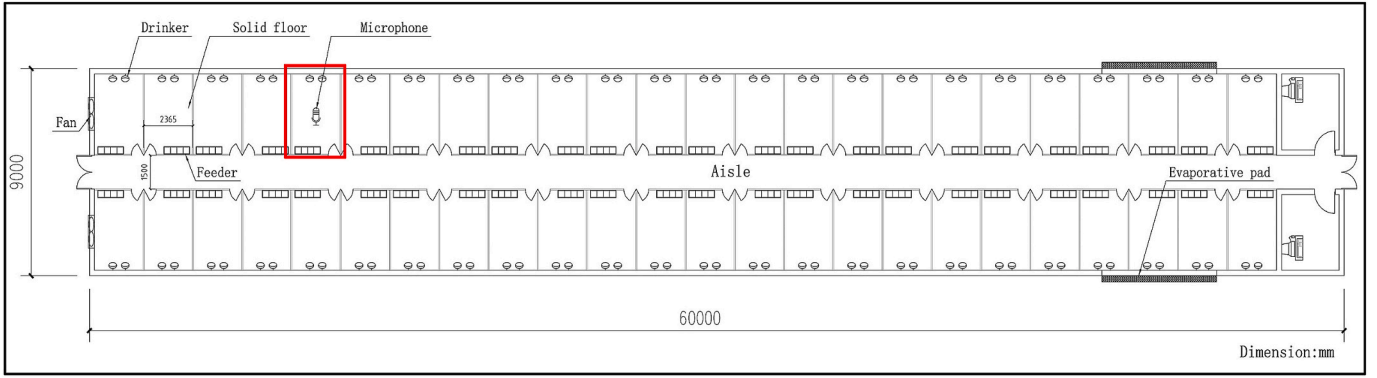


Fig. 1. The experiment setup. According to the layout of the barn No.3, pen No. 10 (Marked Red) was adopted for the experiment, where six pigs were housed.

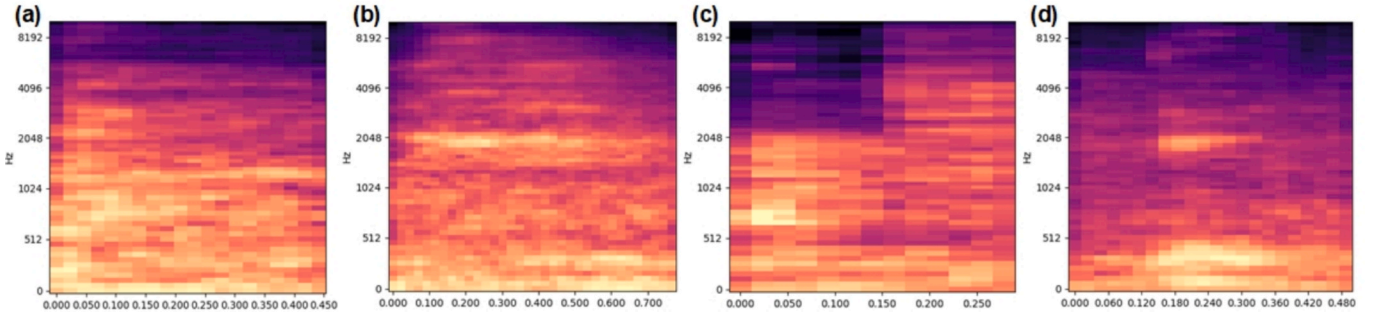


Fig. 2. Log mel-spectrograms of four pig vocalization types, (a) cough, (b) scream, (c) sneezing, (d) other sound.

divided into the training, validation, and test datasets in a ratio of 8:1:1 by randomly selecting samples from each vocalization category (Table 1). In case of overfitting during the training of the recognition model, this study conducted data augmentation on the training dataset. The augmentations used in this study included adding Gaussian noise, pitch shift, time stretch, time mask, and bit reduction, which produced six augmented datasets.

2.2. Multimodal pig audio representation and fusion

The SimSiam framework (Chen and He, 2020) can avoid model collapse and generate meaningful feature representations with only positive pairs and stop-gradient operation, making it convenient to deploy in production conditions. Therefore, this study adopted the SimSiam architecture as the baseline framework. In SimSiam, the inputs are two randomly augmented views x_1 and x_2 from an image x . Then, SimSiam introduces an encoder network f , consisting of a backbone network and a projection multilayer perceptron (MLP), to process the two views by sharing weights between the two views. Next, SimSiam uses a prediction MLP q to transform the output of one view and matches it to the other view. During training, the network weights are updated as follows. x_1 and x_2 are fed to the encoder network f to obtain $z_1 = f(x_1)$ and $z_2 = f(x_2)$. Next, SimSiam uses q to obtain $p_1 = q(f(x_1))$ and $p_2 =$

$q(f(x_2))$. A symmetric loss L is calculated:

$$L = \frac{1}{2}D(p_1, \text{stopgrad}(z_2)) + \frac{1}{2}D(p_2, \text{stopgrad}(z_1)) \quad (1)$$

Where D is the negative cosine similarity, stopgrad is stop-gradient operation. During training, the network weights are updated by minimizing L .

In this study, multimodal pig audio included the raw waveform from the time domain and the log-mel spectrogram and MFCC from the time–frequency domain. In order to process the multimodal pig audio, the SimSiam architecture was extended to the multimodal setting, learning representations from both the time and time–frequency domains (Fig. 3). The MSS used a Transformer for the waveform encoder, thus learning waveform representations by leveraging the powerful abilities in modeling long-term dependencies. Prior to the Transformer encoder, MSS adopted a convolution subsampling layer to reduce the dimension of the input audio waveform, thus reducing the computation load and improving the training efficiency. Regarding the spectral representation encoder, this study used a lightweight SeResNet18 to capture the spatial representations, achieving the balance between model efficiency and accuracy. In the end, the two representations were used to calculate the training loss. In order to enable the MSS to learn more generalized representations, the MSS separately created two views from two different segments, thus increasing the difficulty of identifying the two views as the same sample. For a given audio clip A with a duration of 1.5s, MSS randomly crop two different segments A_1 and A_2 , each with a duration of 1s, respectively, allowing an overlap of at least 50 % between the two segments (Li et al., 2024). This overlap between A_1 and A_2 was to guarantee that the similarity between the two segments was not lost, thereby making it reasonable for the MSS to identify the two segments as the same sample. Then, two data augmentations were applied to the audio segments A_1 and A_2 , respectively, which were randomly selected among adding Gaussian noise, pitch shift, time stretch, time mask, and bit reduction. Next, A_2 was converted from the

Table 1

Statistics of the supervision dataset for the construction of the vocalization recognition model.

Vocalizations	Original dataset	Training	Augmented training	Validation	Test
Cough	3090	2472	17,304	309	309
Scream	3930	3144	22,008	393	393
Sneezing	1110	888	6216	111	111
Other sounds	1370	1096	7672	137	137
Total	9500	7600	53,200	950	950

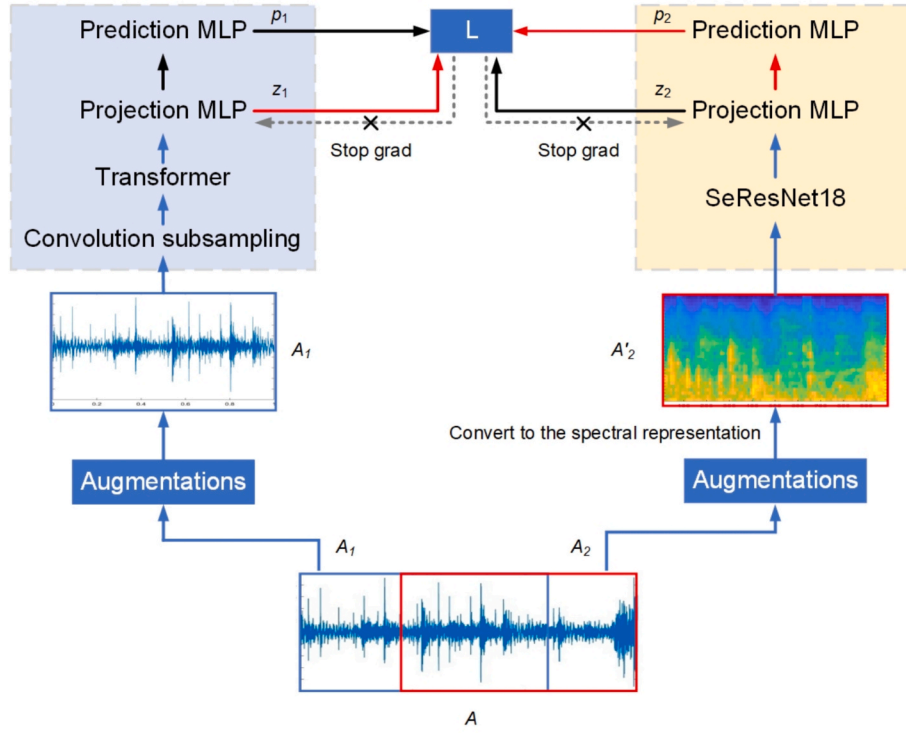


Fig. 3. The proposed multimodal pig audio representation and fusion framework, the symmetric loss L could be computed by feeding the augmented raw waveforms and spectral representations to the Transformer and SeResNet18 encoders, respectively. The prediction MLP shared weights between the two branches. Stop grad indicated the stop gradient operation.

raw waveform to the spectral representation, obtaining A'_2 . By this means, two views of the input audio clip A were created, i.e., (A_1, A'_2) .

The encoding process of MSS was shown in Fig. 3. The convolution subsampling layer had a kernel size of 17 and a stride of 6, which followed by a BN and a ReLU. Next, a Transformer containing two blocks and four heads for each block was used to extract representations from the augmented audio segment A_1 , followed by a projection MLP containing three layers. The dimension of the two hidden linear layers was 512, with each followed by a batch normalization layer (BN) and a rectified linear unit (ReLU) in sequence. The output linear layer had a dimension of 2048, followed by a BN. Similarly, the log-mel spectrogram A'_2 was processed by the SeResNet18 encoder to obtain the spatial representations, which were then fed to a projection MLP with the same architecture as for the audio segment. The prediction MLP of the MSS contained two layers. The dimension of the hidden linear layer was 512, followed by a BN and a ReLU in sequence. The output linear layer, which had a dimension of 2048, did not have BN or ReLU. The loss function employed was the same as in the SimSiam architecture.

2.3. Pig cough recognition

Pig cough recognition could be regarded as a downstream task of the proposed MSS. Therefore, the encoders in the MSS were directly adopted by the pig cough recognition model, thus allowing more powerful feature extraction by leveraging the pre-trained weights obtained by the MSS. As shown in Fig. 4, the proposed pig cough recognition model had a two-stream architecture, consisting of two core blocks. The multimodal audio feature extraction (MFE) block was composed of the two encoders in the MSS framework, and the cross-modal audio feature extraction (CFE) contained two convolutional blocks, i.e., ConvB1 and ConvB2. For an input audio clip I_A , it was first augmented and fed to the MFE, in which the two encoders were employed to process the I_A separately and extract multimodal audio feature representations F_{wave} and F_{spec} that had a size of $(B, 2048)$ (B donated batchsize). Next, the extracted feature representations F_{wave} and F_{spec} were element-wise added to obtain the F_{ws} and fed to the CFE. The CFE block employed the two convolutional blocks. ConvB1 contained three linear layers,

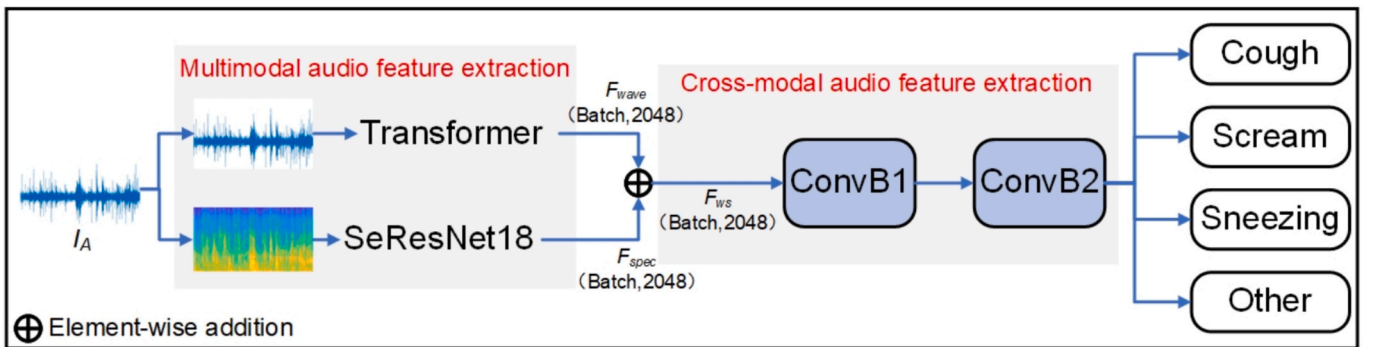


Fig. 4. Architectures of the proposed pig cough recognition model. The model consisted of the multimodal audio feature extraction (MFE) and the cross-modal audio feature extraction (CFE) blocks.

whose dimension was 2048, 512, and 2048, respectively, forming a bottleneck structure. The first two of the linear layers were followed by a BN and a ReLU in sequence, while the output linear layer only had a BN. ConvB2 contained two linear layers. The former one had a dimension of 256 and was followed by a ReLU, while the output linear layer converted the cross-modal feature representations to pig vocalization predictions.

2.4. Performance evaluation

In the evaluation of the MSS framework, this study adopted the standard linear evaluation protocol in the downstream task of pig cough recognition, in which the MFE (the pre-trained encoders) were frozen as a feature extractor, and the CFE (the linear classifier) was trained using the supervision dataset. In order to validate the proposed MSS framework for multimodal pig audio representation and fusion, this study compared the performances of different combinations of pig audio modalities, i.e., raw waveform + log-mel spectrogram (log-mel + wave), raw waveform + MFCC (MFCC + wave), and log-mel spectrogram + MFCC (MFCC + log-mel). In addition, this study also conducted comparisons on the performance of single pig audio modalities in the downstream task using different contrastive learning frameworks. Specifically, the compared models were (1) the raw waveform with multimodal pig audio pre-training by the MSS framework (wave + MSS), (2) the raw waveform with single audio modality pre-training by the WaveBYOL framework (wave + WaveBYOL), (3) the log-mel spectrogram with multimodal pig audio pre-training by the MSS framework (log-mel + MSS), (4) the log-mel spectrogram with single audio modality pre-training by the SimSiam framework (log-mel + SimSiam), (5) the MFCC with multimodal pig audio pre-training by the MSS framework (MFCC + MSS), and (6) the MFCC with single audio modality pre-training by the SimSiam framework (MFCC + SimSiam).

In order to validate the MSS framework for addressing the issue of limited training samples, as well as to understand the limitations, this study experimented with reduced training samples. For each time, only a small portion of samples was randomly selected from each category in the supervision dataset, thus establishing six reduced supervisions, i.e., s_0.5, s_0.4, s_0.3, s_0.2, s_0.1, s_100. The performance of the proposed method was tested over the reduced supervision datasets. Notably, the numbers represented the percentage of samples randomly selected, while s_100 indicated that 100 samples were randomly selected from each category.

In order to show the superiority of the proposed pig cough recognition method, this study conducted comparisons with state-of-the-art pig cough recognition methods on the recognition accuracy and inference time. AST was used as one of the comparison methods since it was proven effective in audio classification and has achieved excellent recognition results of pig abnormal vocalizations in Xie et al. (Xie et al., 2024). In addition, the classifier-wise fusion method (Claf) proposed by Yin et al. (Yin et al., 2023) was also adopted since it shared the same concept with the proposed method of recognizing pig coughs through multimodal audio fusion. In addition, the original Vision Transformer (ViT) (Dosovitskiy et al., 2020) was also adopted as one of the comparison methods. It was worth noting that the above DL-based comparison methods were built upon the supervision dataset since they were supervised. The proposed pig cough recognition method was pre-trained over the pre-training dataset by the MSS framework, following which the pre-trained encoders and the linear classifier are fine-tuned together over the supervision dataset. In addition to the DL-based methods, this study also adopted three ML-based methods for comparison, i.e., Random Forest (RF), Support Vector Machine (SVM), and Back Propagation Neural Network (BPNN). In order to build the ML models, this study extracted the Logarithmic filter bank (LogFbank) features from pig sound recordings as the model input of the ML models. Specifically, the LogFbank, LogFbank Delta (Δ LogFbank), and LogFbank Delta-Delta ($\Delta\Delta$ LogFbank), each containing 40 coefficients, were extracted and concatenated.

This study used confusion matrix to compare the performance of the recognition methods (Ma et al., 2018; Wang et al., 2024). Each column of the confusion matrix corresponded to the actual pig vocalization while each row corresponded to the predicted pig vocalization. Diagonal and off-diagonal values corresponded to correctly and incorrectly classified samples, respectively. According to the confusion matrix, this study calculated four evaluation metrics to evaluate the performances of the recognition models, i.e., Precision, Sensitivity, F1 score, and Accuracy. Detailed calculations were as follow.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

where TP was the number of true positive instances, TN was the number of true negative instances, FP was the number of false positive instances, and FN was the number of false negative instances.

The models proposed in this study were implemented using the PyTorch deep learning framework. The NVIDIA GeForce RTX 3090 (24 GB memory) with CUDA 12.1 was utilized for all the model training. The hyperparameters for the MSS and the pig cough recognition model were summarized in Table 2. During the training process, mix-up augmentation (Kim and Choi, 2023; Niizumi et al., 2023) was adopted, generating new samples by linearly interpolating two random training samples. The linear classifier and the comparison models were training over the training dataset and validated over the validation dataset. The models that achieved the minimum validation loss were adopted and tested over the test dataset.

3. Results

3.1. Results of the multimodal pig audio representation and fusion

The results of different combinations of pig audio modalities, as well as single pig audio modalities, for the downstream task of pig cough recognition were shown in Fig. 5. The multimodal pig audio representation and fusion with the proposed MSS framework improved the accuracy of pig cough recognition, as combining two audio modalities led to improved recognition accuracy compared to using a single audio modality. The MFCC + log-mel achieved the best recognition accuracy of 91.37 %, revealing that most pig vocalization predictions and the true values matched (Table 3). In the performance per category, MFCC + log-mel achieved superior results on category cough and scream compared to category sneezing and other sound. The F1-scores on category cough and scream were 93.40 and 91.86, respectively, followed by category other sound and sneezing with F1-scores of 88.06 and 87.74, respectively. Out of 312 cough predictions, MFCC + log-mel made 290 correct classifications, achieving a precision of 92.95 %. Similar results were

Table 2
Hyperparameters for the models.

Hyperparameters	MSS	Pig cough recognition model
Optimizer	Stochastic gradient descent with a momentum of 0.9	Stochastic gradient descent with a momentum of 0.9
Weight Decay	0.0001	0.05
Learning rate	0.1 followed by a cosine decay schedule with warmup	0.001 followed by a cosine decay schedule with warmup
Batch size	4096	1024
Maximum epoch	2500	1000

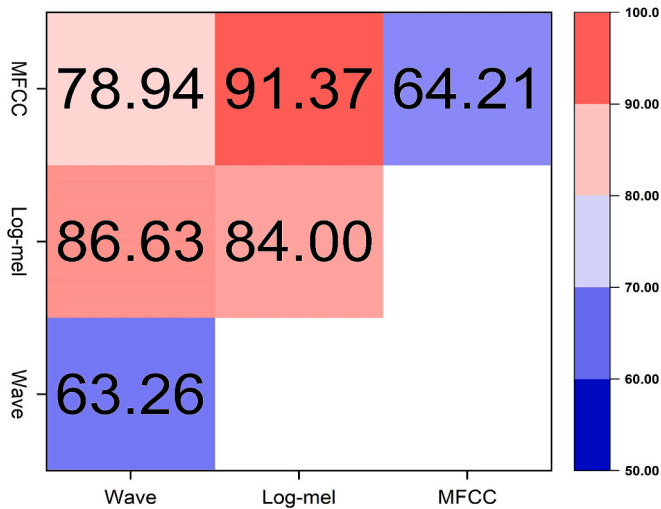


Fig. 5. The linear evaluation results of different combinations of pig audio modalities. Diagonal values represented the results of single modality pre-training, i.e., MFCC only, log-mel only, and raw waveform only.

achieved on category scream, in which 90.39 % of 406 predictions were correct. The precision values for category sneezing and other sound were 92.08 % and 90.08 %, respectively. Regarding the recall, 93.85 % of 309 cough samples were correctly classified, and similar recognition results were observed on the category scream, with 93.38 % of 393 scream samples correctly classified. In contrast, MFCC + log-mel made more errors predicting category sneezing and other sound, with recall values of 83.78 % and 86.13 %, respectively.

The log-mel + wave (Table 4) and MFCC + wave (Table 5) yielded worse results than the MFCC + log-mel, with accuracies of 86.63 % and 78.94 %, respectively. Notably, neither model performed well on category sneezing, and errors primarily came from the misclassification with category cough (Table 4 and 5). In the performance of single modality pre-training-based pig cough recognition, the log-mel spectrogram performed best, achieving an accuracy of 84.00 %, which was even better than the MFCC + wave. The MFCC and raw waveform presented close performance, with accuracies of 64.21 % and 63.26 %, respectively.

The results of using single pig audio modalities in the downstream task across various contrastive learning frameworks were presented in Table 6. It could be observed that all single modalities benefitted from the multimodal pig audio pre-training and achieved better performance compared to those pre-trained with single modalities, proving that the proposed MSS framework for multimodal pig audio representation and

fusion could enhance the generalization ability of single audio modalities. In particular, the combination of MFCC and log-mel spectrograms remarkably promoted the performance of MFCC + MSS, which achieved an accuracy of 90.63 % and increased relatively upon the MFCC + SimSiam by 26.42 %. Meanwhile, the log-mel + MSS was also significantly improved, achieving an accuracy of 90.11 % and increasing upon the log-mel + SimSiam by 6.11 % due to the pre-training with the combination of raw waveforms and log-mel spectrograms. In contrast, multimodal pig audio representation and fusion contributed less significantly to wave + BYOL, with an improvement of 1.58 % resulting from the pre-training with the combination of the raw waveforms and log-mel spectrograms.

3.2. Comparison results with the state-of-the-art methods

The comparison results with the state-of-the-art pig cough recognition methods were presented in Fig. 6. As shown in Fig. 6a, all DL-based models performed well and achieved accuracies over 90 %. Specifically, the MFCC + log-mel + MSS proposed in this study, which was pre-trained and fine-tuned over the pre-training and supervision datasets in sequence (MFCC + log-mel + Fine-tuned, Fine-tuned), achieved the best accuracy of 94.32 %. Following was the MFCC + log-mel that was directly trained from scratch (MFCC + log-mel + LFS, LFS) over the supervision dataset, achieving a slightly inferior accuracy of 94.11 % to the Fine-tuned model. Both Fine-tuned and LFS models outperformed the MFCC + log-mel + MSS pre-trained over the pre-training dataset and linear probing over the supervision dataset in sequence (MFCC + log-mel + Linear, Linear). Similar to the proposed pig cough recognition method, the ClaF adopted multimodal pig audio features, i.e., acoustic and spectral features. However, it failed to yield comparable performance to the pig cough recognition models proposed in this study, with an accuracy of 90.11 % that was even outperformed by the Linear model. The transformer-based recognition models, i.e., ViT and AST, presented slightly worse yet still comparable performance with log-mel spectrograms as model input. The accuracies of the AST and ViT were 92.21 % and 93.47 %, respectively.

As shown in Fig. 6b, the superiority of the DL-based models over ML-based models in recognizing pig cough in practical production conditions could be observed. The ML-based models, i.e., RF, SVM, and BPNN, could not yield comparable performance to the proposed pig cough recognition model, as well as to the other comparison DL-based models, such as AST and ViT. The best-performing ML-based model was the RF, achieving an accuracy of 89.79 %. Following were the SVM and BPNN, with accuracies of 82.95 % and 73.68 %, respectively.

The results in Fig. 6 demonstrated the superiority of the DL-based models to the ML-based models. Therefore, this study mainly focused

Table 3

Confusion matrix of the linear classifier with MFCC + log-mel.

Vocalizations	Cough	Scream	Sneezing	Other	Precision (%)	Recall (%)	F1-score
Cough	290	14	6	2	92.95	93.85	93.40
Scream	15	367	8	16	90.39	93.38	91.86
Sneezing	0	7	93	1	92.08	83.78	87.74
Other	4	5	4	118	90.08	86.13	88.06
Accuracy (%)	91.37						

Table 4

Confusion matrix of the linear classifier with log-mel + wave.

Vocalizations	Cough	Scream	Sneezing	Other	Precision (%)	Recall (%)	F1-score
Cough	284	27	15	7	85.29	91.91	88.47
Scream	16	352	12	18	88.44	89.57	89.00
Sneezing	6	10	79	4	79.80	71.17	75.24
Other	3	4	5	108	90.00	78.83	84.05
Accuracy (%)	86.63						

Table 5

Confusion matrix of the linear classifier with MFCC + wave.

Vocalizations	Cough	Scream	Sneezing	Other	Precision (%)	Recall (%)	F1-score
Cough	243	16	21	5	85.26	78.64	81.82
Scream	43	359	29	30	77.87	91.35	84.07
Sneezing	9	9	42	4	65.63	41.58	50.91
Other	14	9	9	98	75.38	71.53	73.41
Accuracy (%)	78.94						

Table 6

Results of the linear evaluation for different pre-training frameworks.

Pre-training frameworks	Training inputs	Downstream inputs	Downstream test accuracy (%)
MSS	Wave + log-mel	Wave	64.84
WaveBYOL	Wave	Wave	63.26
MSS	Wave + log-mel	Log-mel	90.11
SimSiam	Log-mel	Log-mel	84.00
MSS	MFCC + log-mel	MFCC	90.63
SimSiam	MFCC	MFCC	64.21

on the DL-based models for evaluating the inference time (Table 7). It could be seen that the proposed MFCC + log-mel + MSS demonstrated the best inference efficiency. The inference time over the test dataset, which contained 950 samples, was 11.78s, indicating a samples per second (SPS) of 80.66. The AST and ViT demonstrated similar inference speed since AST was the counterpart of ViT in audio domains. The inference speed of the AST and ViT was 63.84 and 64.11 SPS, respectively. In contrast, the ClaF showed the worst inference speed of 30.76 SPS.

3.3. Results with limited training samples

Since the MFCC + log-mel demonstrated excellent recognition capability for pig coughs with the MSS framework, the Linear model was used to investigate the effects of the limited training samples by testing over the reduced supervision datasets. Given the performance of the AST in the comparison experiments, it was also adopted. As shown in Fig. 7, both the Linear model and AST degraded as the training samples

decreased. However, the difference between the two models was that the performance reduction of the AST was much more rapid than that of the Linear model. When the training samples were sufficient, both the Linear model and AST achieved accurate recognition results, with accuracies of 91.37 % and 92.21 %, respectively. As the number of training samples decreased from 2472 to 100, the performance of the Linear model declined to 80.00 %, with a decrease of 11.67 %. In contrast, the AST was significantly influenced by the lack of train samples, and the performance declined to 49.50 % over the s100, marking a decrease of 42.71 %. Across the reduced supervision datasets, the Linear model consistently outperformed the AST. The results indicated that the proposed MSS framework could mitigate the influence of data scarcity.

4. Discussion

4.1. Potential of multimodal pig audio representation and fusion

This study proposed a novel pig cough recognition method in practical production conditions, in which a multimodal pig audio

Table 7

Inference time of DL-based models over the test dataset. Since the Fine-tuned, LFS and Linear shared the same architecture, this study used MFCC + log-mel + MSS to represent the three models.

Models	Inference time over the test dataset	Samples per second (SPS)
AST	14.88	63.84
ClaF	30.88	30.76
ViT	14.82	64.11
MFCC + log-mel + MSS	11.78	80.66

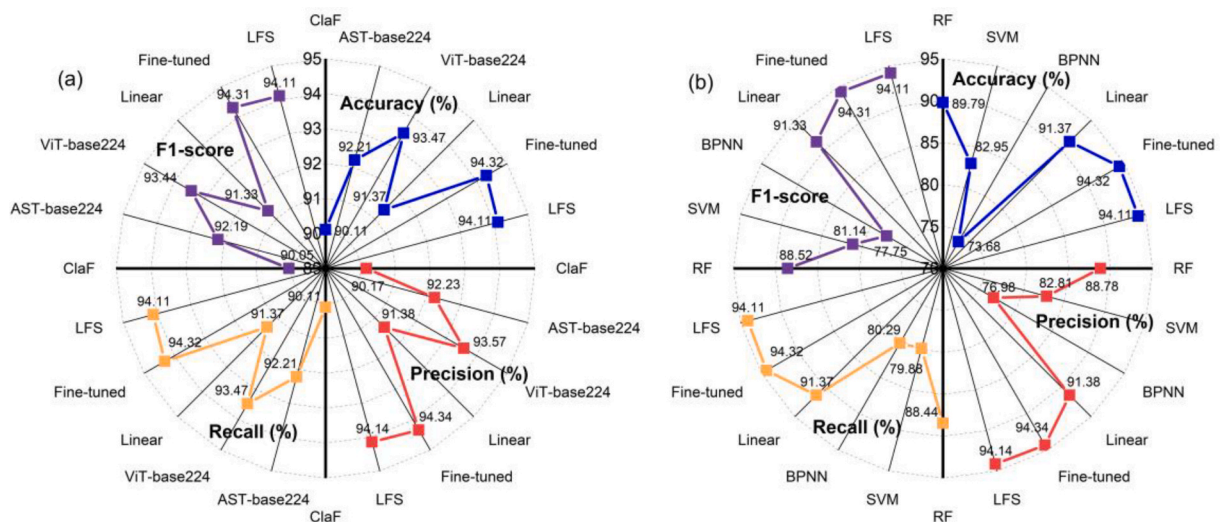


Fig. 6. Comparison results with state of the arts, including comparisons with deep learning-based models (a) and machine learning –based models (b). Fine-tuned indicated the MFCC + log-mel + MSS pre-trained and fine-tuned over the pre-training and supervision datasets in sequence, Linear indicated the MFCC + log-mel + MSS pre-trained over the pre-training dataset and linear probing over the supervision dataset in sequence, LFS indicated the MFCC + log-mel trained from scratch over the supervision dataset.

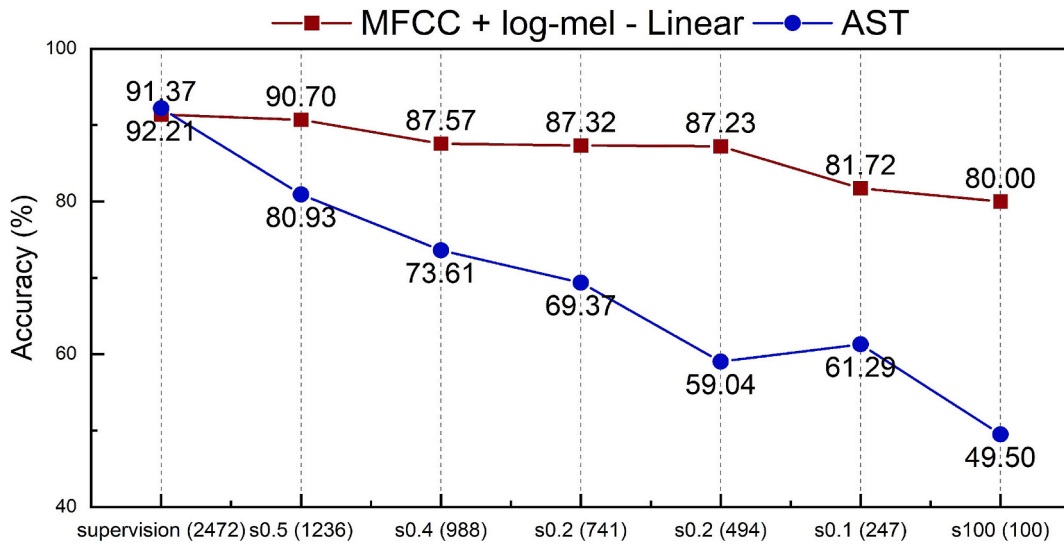


Fig. 7. Performance of the models over the reduced supervision datasets. The numbers in the brackets were the numbers of cough samples in the corresponding datasets, respectively.

representation and fusion (MSS) framework was proposed. Based on the MSS framework, accurate pig cough recognition results were obtained by the combination of the MFCC and log-mel spectrograms, even under the linear evaluation protocol (Fig. 5). Besides, the MSS framework also significantly enhanced the performance of the single modalities in recognizing pig coughs in practical production conditions (Table 6). As shown in Fig. 6a, the proposed model fusing two audio modalities yielded better performance than the unimodal AST and ViT. According to the above results, it could be concluded that using multimodal pig audio could improve the accuracy of pig cough recognition in practical production conditions, agreeing with previous studies on pig cough recognition (Ji et al., 2022; Pann et al., 2024; Shen et al., 2021; Yin et al., 2023). The remarkable performance of the MSS framework might be attributed to the approaches used to create the two views for contrastive learning. This study adopted multimodal pig audio as the input to the MSS framework, generating two views that were semantically related yet different in form. Besides, this study created the two views from two pig audio segments with an overlap of over 50 %. Therefore, it was challenging to identify the two views as the same sample (Li et al., 2024), and left the MSS framework no choice but to learn more generalized representations to achieve accurate results in the downstream task of pig cough recognition. Besides, pig vocalization varied with age, breed, and sex (Upadhyaya et al. 2023b). The adopted approaches for creating the two views for the MSS enabled to learn audio representations were invariant to the slight difference in pig vocalizations, thus allowing the pig cough method to accurately recognize coughs across variations. In summary, the MSS framework boosted the pig cough recognition in practical production conditions.

4.2. Applicable potential of proposed method in practical production conditions

The proposed method had the applicable potential to recognize pig cough in practical conditions. Previous studies conducted audio pre-processing to eliminate noise and improve recognition accuracy, such as noise filtering (Hou et al., 2024; Shen et al., 2022; Wang, 2024; Yin et al., 2023; Zhang et al., 2019) and endpoint detection (Pan et al., 2024; Xie et al., 2024; Zhang et al., 2019). However, these denoising methods might not work well when generalized to other pig barns since the barn acoustic environment might be largely varied due to the difference in breed, age, sex, and the machinery and ambient noises (Upadhyaya et al., 2023b), thus, to some extent, causing the methods to noticeably

deteriorate, which might explain the reasons that the ClaF were not comparable to the proposed pig cough recognition method. In contrast, the proposed pig cough recognition method directly took the raw audio segments as model input, thus eliminating the chance of model deterioration caused by the inaccurate audio denoising and simplifying the inference technological process. Consequently, the proposed pig cough recognition method had great potential to adapt to the changing acoustic environment.

As shown in Table 7, the proposed pig cough recognition method showed advantages over the state-of-the-art methods regarding inference speed, achieving an SPS of 80.66. The reasons for the superior inference speed were two-fold. The MFCC + log-mel + MSS adopted the SeResNet18 as the feature encoder, fully leveraging the inherent characteristics of DL on automatic feature extraction. Besides, the SeResNet18 balanced model efficiency and accuracy, thus making it efficient to learn robust feature representations with few parameters (Ma et al., 2023). In contrast, the AST and ViT had much larger parameters than the proposed MFCC + log-mel + MSS, which could explain the inferior inference speed to the proposed MFCC + log-mel + MSS. In terms of the ClaF, it still needed manual acoustic feature extraction, which occupied most of the inference time and might explain the poor inference speed of 30.76 SPS. Consequently, the proposed pig cough recognition method was more applicable in practical production conditions.

The downstream task dataset could influence the performance of the proposed method in practical production conditions. In the performance per category (Tables 3-5), all models achieved superior results on the category cough and scream compared to the category sneezing and other sound, which could be attributed to the class imbalance of the dataset. In the supervision dataset, the number of samples in the category cough and scream was larger than that in the category sneezing and other sound. Therefore, models demonstrated a bias toward the majority categories, agreeing with the conclusions by Ma et al. (2018) and Wang et al. (2024). Of the four categories, sneezing was the worst performing category, particularly for the MFCC + wave (Table 6). The precision and recall values were 65.63 % and 41.58 %, respectively, indicating that more than half of the sneezing samples were misclassified as category cough and scream. A likely cause for this result might be that the category sneezing was acoustically similar to the category cough and scream, while the MFCC and raw waveform modalities had limited distinguishing ability compared to the log-mel spectrogram (Fig. 5), which also explained the results that both the MFCC + log-mel and the

log mel + wave outperformed the MFCC + wave.

The complex acoustic environments in pig barns posed significant challenges to pig cough recognition (Shen et al. 2022; Wang et al. 2024). According to the barn settings of this study, the pig barn consisted of 48 pens, each housing six growing pigs (Fig. 1), indicating that overlapping pig vocalizations might occur, especially when respiratory disease in pigs became more severe (Wang et al. 2024). When pigs vocalized simultaneously, individual vocalizations blended, making it difficult to separate and recognize distinct vocalizations. Consequently, the pig cough recognition methods trained using clean vocalization samples would exhibit performance deterioration due to the lack of corresponding training data. In the built supervision dataset, the duration of pig vocalizations, such as cough, scream, and sneezing, was short (mostly less than 0.5 s), resulting in relatively few instances of overlapping pig vocalizations. Consequently, this study excluded pig vocalization samples with heavy overlaps during labeling. In order to further enhance the practical applicability of the proposed method, overlapping pig vocalizations, in addition to acoustically similar vocalizations and background noises, should also be taken into consideration (Wang et al. 2024).

4.3. Contribution of the MSS on addressing the issue of limited training samples

The proposed MSS framework addressed the issue of limited training samples for recognizing pig coughs in practical production conditions. As shown in Fig. 7, the proposed Linear model maintained robustness over the reduced supervision datasets. Remarkably, the Linear model effectively maintained its accuracy even with a 50 % reduction in training samples compared to the supervised dataset. Although the performance degraded as training samples decreased, it still yielded decent recognition accuracy. This robustness of the Linear model was believed to be rooted in the proposed MSS framework, which leveraged a large amount of unlabeled data to generate supervisory signals for the pig cough recognition model. In contrast, the performance of the AST model over the reduced supervision datasets degraded rapidly compared to that over the supervision dataset (Fig. 7). Specifically, an 11.28 % performance reduction was observed when the number of training samples was halved. Furthermore, the performance of the AST kept deteriorating rapidly with the decrease in the number of training samples. A possible explanation for the rapid deterioration of the AST model might be that the state-of-the-art DL models were data-intensive, thus requiring a large amount of labeled training samples for accurate recognition results. Consequently, the performance significantly decreased when the labeled training samples were limited (Shen et al., 2022). This study revealed that self-supervised pre-training was less applied to adequate labeled training samples, agreeing with Zhang et al. (Zhang et al., 2024). As shown in Fig. 5, the Fine-tuned model achieved the best recognition results over the supervision dataset and slightly outperformed the LFS model. Both the Fine-tuned and LFS models, as well as the ViT and AST, outperformed the Linear model, indicating that the contributions of the proposed MSS framework to the pig cough recognition were limited when the labeled training samples were sufficient. On the contrary, the proposed MSS framework progressively enhanced the performance of the pig cough recognition as the number of labeled training samples decreased in the reduced supervision datasets. In particular, when the number of cough samples was reduced to 100, the Linear model still achieved an accuracy of 80 %, which was far superior to the AST model, with an accuracy of 49.50 %. Therefore, it could be concluded that the proposed MSS framework made a promising tool for pig cough recognition in practical production conditions where the labeled cough samples were severely insufficient.

4.4. Limitations and future work

In this study, a novel MSS framework was proposed, based on which

the log-mel spectrogram and MFCC were fused to recognize pig cough in practical production conditions. Under linear evaluation protocols, the Linear model achieved an accuracy of 91.37 % in practical production conditions. Compared to the full fine-tuning model, the recognition accuracy of the Linear model is expected to be further improved. In future work, parameter-efficient fine-tuning (PEFT) methods, such as prompt tuning and adapter tuning, will be explored to further enhance model performance in practical production conditions. In addition, more sound data under different pigsties and background noises, including overlapped vocalization samples will be collected to test the generalization ability of the pig cough recognition model, as well as to further improve the model applicability in practical production conditions. In the meantime, the proposed method will be generalized to various livestock breeds, including calves and laying hens.

5. Conclusion

This study proposed a novel MSS framework, based on which a recognition method for pig coughs in practical production conditions was proposed. The results demonstrated that fusing multimodal pig audio could improve the performance of pig cough recognition in practical production conditions, and the combination of MFCC and log-mel spectrogram achieved the best results in the linear evaluations. After fine-tuning the supervision dataset, the proposed MFCC + log-mel achieved superior results to the state-of-the-art supervised pig cough recognition methods. The proposed MSS framework demonstrated the outstanding capability of learning pig audio representations compared to the state-of-the-art contrastive learning frameworks, and could enhance the representation ability of single pig audio modalities. In addition, the proposed MSS framework effectively addressed the issue that the labeled training samples were limited in practical production conditions, thus making a necessary supplement to the supervised methods. Given the feasibility of the proposed pig cough recognition method in practical production conditions, future work will extend the methods to inspection robots, thus advancing the early warning of pig respiratory diseases.

CRedit authorship contribution statement

Juncheng Ma: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Zishuo Li:** Validation, Data curation. **Hao Wang:** Resources, Project administration. **Xiao Yang:** Writing – review & editing, Validation. **Dan Xu:** Validation, Software. **Chaoyuan Wang:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The Authors wish to acknowledge the National Science and Technology Major Project (2021ZD0113801), National Center of Technology Innovation for Pigs (NCTIP-XD/B15), the 2115 Talent Development Program of China Agricultural University and Chinese Universities Scientific Fund (2025RC031) for the funding support of this research.

Data availability

Data will be made available on request.

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